



Higher diabetes genetic load in proliferative diabetic retinopathy in South India: The South Indian GeNetics of DiAbetic Retinopathy (SIGNATR) study

A. N. Rizza^{1,2} · Penelope Benckek³ · Rehana Khan^{2,4} · Renee Liu⁵ · E. Ricky Chan³ · Ashley Li⁵ et al. [full author details at the end of the article]

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Abstract

Purpose To identify genomic risk factors for diabetic retinopathy (DR), proliferative diabetic retinopathy (PDR) and diabetic macular edema (DME), in South Indians.

Methods Phenotyping, including optical coherence tomography (OCT), of South Indians ($n = 2538$) with type 2 diabetes (T2D) was obtained. Genome-wide association studies (GWAS) were performed for DR, PDR, and DME with covariate adjustment. The results were replicated in two cohorts. T2D polygenic risk scores (PRS) were examined to determine if DR cases were enriched for T2D loci.

Results A novel locus on chromosome 10 [rs11199996 ($OR = 4.10$, $P = 2.88 \times 10^{-8}$)] was associated with PDR, replicated in other cohorts, and is located near genes involved in retinal light transduction and growth hormone signaling. Several near genome-wide significant loci were identified, including rs76323047 in the glucokinase gene (*GCK*, $P = 3.89 \times 10^{-7}$), a glucose sensor; this risk variant displays higher frequency in Asian populations. T2D PRS showed consistent associations with DR, particularly PDR, suggesting that DR cases have a higher genetic load for T2D.

Conclusion This first DR GWAS in South Indians identified a significant association between a chromosome 10 variant and PDR. We also found notable associations of T2D PRS with DR.

Key messages

What is known

- Established risk factors for diabetic retinopathy (DR) are poor glycemic control and longer duration of diabetes, but they do not fully explain the development and progression of DR.
- Most genetic studies based on DR have been focused on European and East Asian populations, with very limited work in South Indian cohorts, despite their higher disease burden.

What is new

- A novel locus on chromosome 10 located near genes involved in retinal light transduction was associated with proliferative diabetic retinopathy in this South Indian population.
- A near genome-wide significant locus was also identified in the glucokinase gene; the risk variant at this locus has a higher frequency in Asian populations.
- Type 2 diabetes polygenic risk scores showed consistent associations with DR, suggesting that DR cases have a higher genetic load for type 2 diabetes risk variants.

Keywords Genome-wide association study · Genetic risk factors · South Indian · Diabetic retinopathy · Proliferative diabetic retinopathy · Diabetic macular edema

A. N. Rizza and Penelope Benckek contributed equally to the work.

Sinnakaruppan Mathavan, Lucia Sobrin and Sudha K. Iyengar contributed equally to the work.

Introduction

About 74.2 million people have type 2 diabetes (T2D) in India, and Indians exhibit a unique metabolic profile, characterized by higher intra-abdominal fat and insulin resistance, predisposing them to earlier T2D onset and increased susceptibility to T2D complications, including diabetic retinopathy (DR) compared to Western populations [1, 2]. Within India, the prevalence of diabetes-related complications, including DR, is higher in South India compared with other regions [2–9]; for example, DR prevalence was 22.65% in South India vs. 12.27% and 14.15% in Central and Northeast India, respectively [9].

Poor glycemic control and longer duration of diabetes (DOD) are primary DR risk factors but alone do not fully account for DR risk. While additional lifestyle factors may contribute, there is evidence that genetic predisposition plays a key role [10–12]. To date, most DR genome-wide association studies (GWAS) have focused on Europeans and East Asians, potentially overlooking genetic risk factors unique to South Indians [12–16]. The South Indian population, with its high prevalence of T2D and DR, provides an opportunity to understand DR genetic risk factors, including population-specific ones.

Among the forms of DR, proliferative diabetic retinopathy (PDR) and diabetic macular edema (DME) are the major causes of vision impairment [17]. Few genetic studies have focused on these vision-threatening DR types [18]. To our best knowledge, no previous GWAS for DR has incorporated advanced DR phenotyping with optical coherence tomography (OCT) and OCT angiography (OCTA). Identifying genetic associations with PDR, DME and imaging endpoints could provide additional insights.

Beyond individual genetic variants, polygenic risk scores (PRS) have emerged as an important tool for assessing genetic susceptibility to T2D and its complications [19]. Evidence suggests that individuals with a higher T2D genetic burden may also have increased DR risk [20]. Understanding how PRS correlate with DR severity could improve early identification of high-risk individuals.

The primary purpose of the South Indian GeNetics of DiAbetic Retinopathy (SIGNATR) Study [21] is to identify genomic risk factors for DR.

Participants

Two groups were included in the SIGNATR Study [21]: those enrolled in the Sankara Nethralaya Diabetic Retinopathy and Molecular Genetics Study (SN-DREAMS) [4, 22] and those enrolled prospectively at Sankara Nethralaya Hospital. A total of 1,363 individuals were recruited in the

SIGNATR Study, while 1,175 participants were included from the SN-DREAMS Study. Inclusion/exclusion criteria are detailed in Online Resource 1. Briefly, inclusion criteria were South Indian ancestry and T2D. The exclusion criterion was the presence of ocular disease that impeded grading of DR.

Materials and methods

Data were collected as previously described [21] and included DOD and HbA1c. Participants' retinas were imaged with fundus photography and OCT [21]. Some patients also underwent macular OCTA using the Cirrus HD-OCT 5000 (6 × 6 mm area).

Fundus images were graded using the International Clinical Diabetic Retinopathy Disease Severity Scale: no DR, mild non-proliferative diabetic retinopathy (NPDR), moderate NPDR, severe NPDR, or PDR [21, 23]. Of note, for a small minority of the SN-DREAMS participants, the fundus photographs were only graded as having DR or no DR; the presence of DR was not further broken down into levels of NPDR and PDR. There were two imaging-derived measures of DME: clinically significant macular edema (CSME) from photographs and center-involved DME (ciDME) from OCT [21]. OCT central subfield thickness (CST) was recorded. Automated OCTA measures of superficial layer vessel density (VD) and foveal avascular zone (FAZ) were obtained with AngioPlex Metrix (Carl Zeiss Meditec, Dublin, CA). For dichotomous outcomes, the grade from the eye with most advanced DR, CSME, and ciDME was included for each participant. For the continuous outcomes, the value from the eye with the worse outcome (highest CST, lowest VD, and largest FAZ) was included.

Details of the DNA extraction, genotyping, sample and variant quality control (QC), and imputation are included in the Online Resource 1 and Online Resource 2-Supplementary Figs. 1–2. After QC, 2,261 participants remained.

The primary outcome examined was DR vs. no DR. Secondary analyses compared: PDR vs. no DR, PDR vs. no PDR (includes NPDR and no DR), CSME vs. no CSME, ciDME vs. no ciDME, as well as the continuous outcomes of CST, VD, and FAZ. Covariate adjustments were made for HbA1c levels, DOD, DR severity (for the DME analyses) and significantly associated principal components. GWAS were run on well-imputed single nucleotide polymorphisms (SNPs) with minor allele frequency $\geq 1\%$ using the penalized quasi-likelihood approximation to the generalized linear mixed model in R (Genesis package).

Genome-wide significant ($P < 5 \times 10^{-8}$) and suggestive findings ($P < 1 \times 10^{-5}$) from the discovery GWAS analyses were evaluated for replication in existing DR GWAS cohorts

Fig. 1 Manhattan plot of the genome-wide association analysis comparing patients with PDR to those without any DR

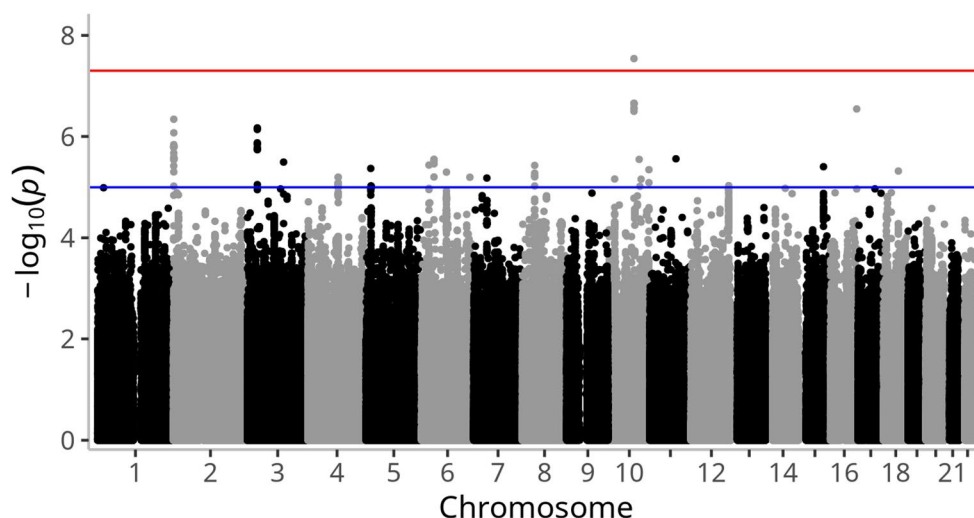


Table 1 Genome-wide significant findings from discovery genome-wide association study

Phenotype	rsID	Chromosome: Position: Allele	Effect allele	Frequency effect allele	P-value	Odds ratio	95% Confidence interval	Nearest gene	Function
PDR vs no DR	rs11199996	chr10:83696925:G:A	A	0.046	2.88×10^{-8}	4.10	[2.51, 6.69]	<i>RP11-344L13.2</i>	Intergenic
PDR vs no PDR	rs75370175	chr2:1032210:C:T	T	0.021	2.79×10^{-8}	7.03	[3.54, 13.96]	<i>SNTG2</i>	Intronic

PDR Proliferative diabetic retinopathy, DR Diabetic retinopathy

using the rma function (R, metafor) for fixed-effects meta-analysis with a linear (mixed-effects) model framework. Replication cohort details are in Online Resource 3-Supplementary Table 1. For the DR severity and OCT-based DME outcomes, replication was available in South Asians [24]. For the CSME analyses, replication was available in both South Asians [24] and Europeans [18].

T2D PRS were calculated from three T2D studies and three glycemic trait (insulin sensitivity, proinsulin, and random glucose) studies across three DR binary outcomes (DR vs. no DR, PDR vs. no DR and PDR vs. no PDR) [25–29]. The PRS summary statistic source populations were South Asian and European [25, 27–30]. PRS method details are in Online Resource 1.

Results

Primary GWAS

The primary GWAS compared participants with any DR ($n = 737$) to those with no DR ($n = 1524$). Online Resource 2-Supplementary Figs. 3 and 4 show the quantile–quantile and Manhattan plots, respectively. There were no genome-wide significant findings. The most strongly associated SNP was on chromosome 13 (rs78193912) with an odds ratio (OR)=0.66 and $P = 1.01 \times 10^{-7}$ (Online Resource 3-Supplementary Table 2).

Secondary GWAS

PDR

In comparing participants with PDR ($n = 289$) to those without DR ($n = 1524$), there was one genome-wide significant chromosome 10 locus with lead SNP rs11199996 (OR=4.10, $P = 2.88 \times 10^{-8}$) (Fig. 1, Table 1, and Online Resource 3-Supplementary Table 3).

In comparing participants with PDR ($n = 289$) to those without PDR ($n = 1906$), there was one chromosome 2 genome-wide significant locus with lead SNP rs75370175 (OR=7.03, $P = 2.79 \times 10^{-8}$) (Fig. 2, Table 1, and Online Resource 3-Supplementary Table 4). The sample size of the PDR vs. no PDR analysis was smaller than the DR vs. no DR because there were SN-DREAMS cases where it was possible to determine DR vs. no DR status given the documentation available, but it was not always possible to determine PDR vs. NPDR status.

DME

For both DME comparisons [(CSME ($n = 245$) vs. no CSME ($n = 1597$) and ciDME ($n = 81$) vs. no ciDME ($n = 304$)], there were no genome-wide significant findings. The most strongly associated SNPs were on chromosome 4, rs12640221 (OR=0.52, $P = 2.11 \times 10^{-6}$) and chromosome 2, rs115903989 (OR=38.47, $P = 2.44 \times 10^{-7}$), respectively

Fig. 2 Manhattan plot of the genome-wide association analysis comparing patients with PDR to those without PDR

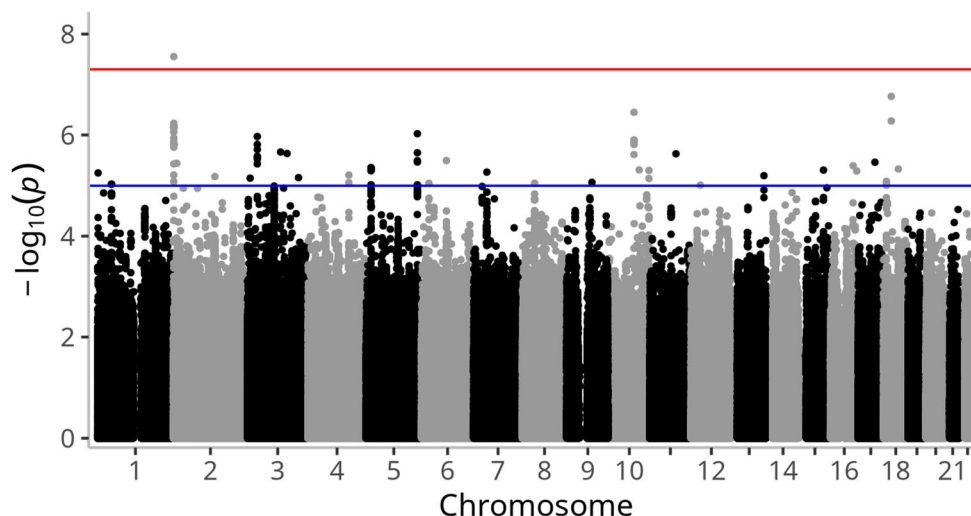


Table 2 Fixed-effects replication meta-analysis genome-wide significant findings

rsID	Discovery <i>P</i> -value	Singapore cohort <i>P</i> -value	Phenotype	Effect allele	Direction of effect	Meta-analysis <i>P</i> -value	Meta-analysis Odds ratio	Meta-analysis 95% Confidence interval
rs11199996	2.88×10^{-8}	0.083	PDR vs. no diabetic retinopathy	A	++	1.88×10^{-8}	3.4	[2.2, 5.12]
rs11199996	3.55×10^{-8}	0.027	PDR vs. no PDR	A	++	4.39×10^{-8}	3.0	[2.05, 4.55]

PDR Proliferative diabetic retinopathy

(Online Resource 2-Supplementary Figs. 5–6 and Online Resource 3-Supplementary Tables 5–6). In the CST analysis, the SNP with the lowest *P* value was in the glucokinase (*GCK*) gene (rs76323047) with a $\beta = -1.2 \times 10^{-6}$ and $P = 3.89 \times 10^{-7}$ (Online Resource 2-Supplementary Fig. 7 and Online Resource 3-Supplementary Table 7). This *GCK* variant was also nominally associated with ciDME and CSME (Online Resource 3-Supplementary Table 8). In the VD and FAZ analyses, there were no genome-wide significant findings (Online Resource 1).

Replication

In replication, one locus achieved genome-wide significance (Table 2). The chromosome 10 variant was associated with PDR (both vs. no DR and vs. no PDR) ($P = 1.88 \times 10^{-8}$ and 4.39×10^{-8} , respectively). Online Resources 4 and 5 show all results for SNPs present in the replication cohorts for fixed-effects and random-effects meta-analyses, respectively. We also examined genome-wide significant findings from our previous European and African American GWAS [16]. Those results are in Online Resource 1 and Online Resource 3-Supplementary Table 13 [16].

T2D PRS results

The European-based T2D PRS was consistently associated with PDR and DR (Table 3). The South Asian-based T2D PRS was strongly associated with PDR. Two of three South Asian T2D PRS were also associated with any DR (Table 3). We did not find any association between glycemic trait PRS and DR (Online Resource 3-Supplementary Tables 14 and 15).

Discussion

In this first South Indian DR GWAS, a chromosome 10 locus was significantly associated with PDR. This locus replicated in an independent Singaporean cohort and was associated with PDR, both when comparing with no PDR and with no DR. In addition, T2D PRS were consistently associated with DR and PDR. Finally, a suggestive *GCK* variant was associated with CST. This T2D-risk increasing *GCK* also showed nominal association with other DME outcomes – ciDME and CSME. This consistent association suggests that this variant may be important for DME pathogenesis and potentially serve as a future therapeutic target, as *GCK* activators are in clinical trials [31, 32].

Table 3 Type 2 diabetes (T2D) polygenic risk scores (PRS) and diabetic retinopathy (DR)

Study in which T2D PRS was created	DR Phenotype	Threshold	<i>r</i> -squared	<i>P</i> -value	Odds ratio	95% Confidence interval
European-based PRS						
Mahajan et al	DR vs. no DR	0.001	0.0016	0.0320	1.11	[1.00, 1.23]
Mahajan et al	PDR vs. no PDR	0.001	0.0034	0.0154	1.18	[1.03, 1.34]
Mahajan et al	PDR vs. no DR	0.001	0.0043	0.0090	1.21	[1.05, 1.39]
Suzuki et al	DR vs. no DR	0.001	0.0044	0.0004	1.19	[1.08, 1.31]
Suzuki et al	PDR vs. no PDR	0.001	0.0027	0.0307	1.16	[1.01, 1.32]
Suzuki et al	PDR vs. no DR	0.001	0.0048	0.0062	1.22	[1.06, 1.41]
South Asian-based PRS						
Loh et al	PDR vs. no PDR	0.001	0.0032	1.86E-02	1.17	[1.03, 1.33]
Loh et al	PDR vs. no PDR	0.001	0.0076	3.10E-04	1.27	[1.12, 1.45]
Mahajan et al	PDR vs. no PDR	0.001	0.0084	1.66E-04	1.29	[1.13, 1.47]
Loh et al	PDR vs. no DR	0.001	0.0033	2.26E-02	1.17	[1.02, 1.35]
Loh et al	PDR vs. no DR	0.001	0.0079	4.13E-04	1.28	[1.12, 1.47]
Mahajan et al	PDR vs. no DR	0.001	0.0087	2.13E-04	1.31	[1.13, 1.50]
Loh et al	DR vs. no DR	0.001	0.0012	6.80E-02	1.09	[0.99, 1.20]
Loh et al	DR vs. no DR	0.001	0.0020	1.65E-02	1.12	[1.02, 1.24]
Mahajan et al	DR vs. no DR	0.001	0.0017	2.62E-02	1.12	[1.01, 1.23]

PDR Proliferative diabetic retinopathy

The lead SNP on chromosome 10 (rs11199996) is located 442,557 bases from growth hormone inducible transmembrane protein (*GHITM*). Human growth hormone has important roles in angiogenesis [33, 34]. *GHITM* is, in turn, 75,925 bases from the retinal G protein-coupled receptor (*RGR*) gene, a photoisomerase that catalyzes the conversion of all-trans-retinal to 11-cis-retinal [35, 36]. The protein is exclusively expressed in the retinal pigment epithelium and Mueller cells. The gene is also associated with retinitis pigmentosa [37]. *RGR* may be the target gene for the effects of the rs11199996 variant. This variant has not been identified as being associated with DR by GWAS in other populations. *RGR* may be specific to Asian cohorts because this allele is only present in South and East Asians and in those of African ancestry but is absent in Europeans. This region will need further fine mapping to fully resolve how rs11199996 or a nearby marker in high linkage disequilibrium impacts PDR risk. From the meta-analysis results, for each copy of the A allele at rs11199996, a participant's risk of developing PDR increased approximately 3-fold (Table 2). It is important to note that the association result for rs11199996 in the Singaporean cohort did not reach statistical significance in that cohort alone; it only reached significance in meta-analysis across cohorts. The consistency in the direction of effect across cohorts, however, lends evidence for this association. Also, the allele being of low frequency further limits statistical power for replication in a small cohort and makes it hard to achieve standalone significance. In addition to the locus on chromosome 10, a significant locus on chromosome 2 (rs75370175) near the *SNTG2* gene was identified in the PDR vs. no PDR analysis. *SNTG2* codes for syntrophin gamma 2. It is involved in cell signaling and interactions

with dystrophin-associated proteins, thus it may influence retinal integrity and vascular biology.

Associations between T2D risk variants and DR have been previously found [16, 20]. A multi-ethnic DR GWAS identified a correlation between T2D risk variant load and DR [16]. In the Million Veterans Program (MVP), the European T2D PRS was associated with DR [20]. We used both European and South Asian PRS and found associations with both. The MVP T2D PRS was calculated by PRS deciles, and the OR for DR ranged from 1.08 for the 10–20% T2D PRS decile to 1.59 for the 90–100% decile. This agrees well with our ORs which ranged from 1.11–1.19 for DR. To our knowledge, this is the first South Asian-specific DR PRS. It is also the first testing of a T2D PRS for PDR specifically. The ORs for the T2D PRS were generally higher for PDR compared with DR. The consistency of PRS derived in two different populations, and across different studies, DR outcomes, and populations suggests that a main driver of DR genetic risk may be the genetic risk factors underlying T2D itself.

Our *GCK* variant results also support that metabolic and inflammatory pathways involved in T2D contribute to DR. Others have found that *GCK* mutation carriers are at increased risk of T2D complications [38]. The *GCK* marker shows a higher frequency in Asian populations (G allele frequency is 18% in South Asians and 17% in East Asians, versus 12% in Europeans). *GCK* mutations cause young-onset diabetes [39] and suggest that a *GCK* common variant could be a driver for the younger T2D onset in Indians. Various papers suggest that treatment is not necessary for Maturity-Onset Diabetes of the Young (MODY) caused by *GCK* mutations [40–42], but the potential increased DME

risk may mean that even the *GCK* form of MODY could benefit from treatment in some cases.

One novelty of our study is that we examined OCT-based outcomes of DME: *ciDME* and *CST*. OCT provides more granular assessment of DME. While we did not find any significant DME associations in this modestly-size dataset, continued investigation of this outcome is warranted as larger datasets become available. Another novelty is that we examined OCTA-derived vascular measures. To our best knowledge, this has not been previously examined. Although we did not find any genome-wide associated findings, perhaps due in part to limited sample size, this is an important area for further development. OCTA retinal vascularity measures may be an important endophenotype in DR.

Our findings have clinical implications for improving risk prediction and personalized management of DR. The novel chromosome 10 locus associated with PDR provides a potential target for biomarker development, allowing for earlier identification of individuals at high risk for PDR. The strong association between T2D PRS and both DR and PDR suggest that incorporating genetic risk assessment into diabetes care could help stratify patients based on likelihood of developing vision-threatening complications. This could enable more targeted screening programs, ensuring that high-risk individuals undergo regular retinal examinations and earlier interventions. Additionally, the association of a *GCK* variant with DME raises the possibility that genetic factors influencing glucose metabolism may also contribute to DR severity, which could lead to refined approaches that identify patients who may benefit from specific metabolic interventions. The integration of genetic data with advanced imaging (OCT, OCTA) could further enhance predictive models for DR progression, bridging the gap between genetic risk and structural retinal changes.

There are some limitations to our study. The sample size is modest; in particular, for the OCT and OCTA imaging outcomes, we had limited numbers as every participant was not imaged on these modalities. A small sample size limits our ability to detect variants of modest or small effect sizes. It is possible that there are indeed genetic loci that impact these phenotypes, but the current study does not have sufficient power to detect them. The small sample of the replication cohorts, particularly for advanced forms of DR such as PDR, also limited our ability to replicate findings. Therefore, there is a need for investigation of these findings in larger, independent cohorts including biobanks as well as investigation of our findings with functional studies. While there were no systematic differences in age between the participants with OCTA versus fundus-photography only (Mann–Whitney U test p -value = 0.4897), more recently collected patients who underwent OCTA imaging had a

poorer glycemic index (median HbA1c = 8.6) compared to the group of patients who underwent fundus imaging (median HbA1c = 8.23). This bias may reflect the time-period of data collection; fundus photos were collected in SN-DREAMS while the newer collection relied on OCT and OCTA. This was a cross-sectional study; therefore, we did not have longitudinal imaging. It is possible that the participants had DME in the past that was treated or that DME would occur in the future, therefore leading to DME case/control misclassification. Similarly, for DR severity, it is possible that participants who had no DR or NPDR when enrolled might progress later to more severe DR, resulting in misclassification. However, we did control for DOD to address this limitation. Another limitation is that for a small minority of the SN-DREAMS participants, the fundus photographs were only graded as having DR or no DR without further granularity regarding NPDR level or PDR presence. Since we could not be sure if these participants had PDR or not, we only included them in the DR vs. no DR analysis. They could not be included in the PDR vs. no PDR analysis, hence decreasing the sample size and power of that analysis slightly.

In summary, we present the first South Indian DR GWAS. We found one genome-wide significant variant on chromosome 10 for PDR; this variant lies near genes implicated in retinal light transduction (*RGR*) and growth hormone physiology (*GHITM*). In addition, we found that T2D PRS are associated with both DR and PDR. Future studies should concentrate on larger datasets of South Asian patients with detailed imaging to improve the power to detect novel variants and perhaps consider Indian-specific PRS to identify those at high risk for DR. These efforts could lead to incorporating genetic risk assessment into clinical decision-making, ultimately paving the way for precision medicine approaches in DR management that go beyond modulating traditional risk factors like HbA1c.

Appendix 1: SIGNATR Study Group

A. N. Rizza,^{1,2†} Penelope Benckek,^{3†} Rehana Khan,^{2,4} Renee Liu,⁵ **Kim Brustoski**,³ E. Ricky Chan,³ Ashley Li,⁵ Gayatri Susarla,⁵ **Janine Yang**,⁵ Sam Han,⁵ Katie Huynh,⁵ **Ines Lains**,⁵ Ching-Yu Cheng,^{6,7,8,9} Hengtong Li,^{7,8} Tien Yin Wong,^{6,10} Jaime E Craig,¹¹ Bennet J. McComish,¹² Rajya L. Gurung,¹² Kathryn Burdon,¹² **Vikas Khetan**,¹³ **Sarangapani Sripriya**,² Rajiv Raman,² Sinnakaruppan Mathavan,^{2,14*} Lucia Sobrin,^{5*} Sudha K. Iyengar^{3*}

1. Department of Biotechnology, Alagappa University, Karaikudi- 630 003, India

2. Vision Research Foundation, Sankara Nethralaya, Chennai, Tamil Nadu, India

3. Department of Population and Quantitative Health Sciences, Case Western Reserve University School of Medicine, Cleveland, OH, USA

4. School of Optometry and Vision Science, University of New South Wales, Sydney, Australia

5. Department of Ophthalmology, Harvard Medical School, Massachusetts Eye and Ear Infirmary, Boston, MA, USA

6. Singapore Eye Research Institute, Singapore National Eye Centre, Republic Singapore

7. Centre for Innovation and Precision Eye Health, Yong Loo Lin School of Medicine, National University of Singapore, Singapore, Republic of Singapore

8. Department of Ophthalmology, Yong Loo Lin School of Medicine, National University of Singapore, Singapore, Republic of Singapore

9. Ophthalmology and Visual Sciences Academic Clinical Program (Eye ACP), Duke-NUS Medical School, Singapore, Republic of Singapore

10. Beijing Visual Science and Translational Eye Research Institute, Beijing Tsinghua Changgung Hospital, Tsinghua Medicine, Tsinghua University, Beijing, China

11. Department of Ophthalmology, Flinders Medical and Health Research Institute, Flinders University, Adelaide, SA, Australia

12. Menzies Institute for Medical Research, University of Tasmania, Australia

13. Department of Ophthalmology, Flaum Eye Institute, Rochester, New York, United States

14. MedGenome Labs, Bangalore-560100

Bolded authors names indicate those authors that are included in the title page under the SIGNATR Study Group authorship banner.

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Data availability The dataset is available on dbGAP (Study Accession Number: phs002116.v3.p2).

Declarations

Ethics approval This study was approved by the institutional review boards of the Vision Research Foundation, Case Western Reserve University, and Mass General Brigham. All participants provided informed consent, and the research adhered to the tenets of the Declaration of Helsinki.

Consent to participate Informed consent was obtained from all individual participants included in the study.

Consent to publish Patients signed informed consent regarding publishing their data and photographs.

Competing interests **Financial Interests:** A. N. Rizza receives financial support from the Department of Biotechnology, Government of India to fund this current manuscript.

Penelope Benckek owns stock in Crispr Therapeutics.

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Authors and Affiliations

A. N. Rizza^{1,2} · Penelope Benchek³ · Rehana Khan^{2,4} · Renee Liu⁵ · E. Ricky Chan³ · Ashley Li⁵ · Gayatri Susarla⁵ · Sam Han⁵ · Katie Huynh⁵ · Ching-Yu Cheng^{6,7,8,9} · Hengtong Li^{7,8} · Tien Yin Wong^{6,10} · Jaime E. Craig¹¹ · Bennet J. McComish¹² · Rajya L. Gurung¹² · Kathryn Burdon¹² · Rajiv Raman² · Sinnakaruppan Mathavan^{2,13} · Lucia Sobrin⁵ · Sudha K. Iyengar³  · The SIGNATR study group

✉ Sudha K. Iyengar
ski@case.edu

¹ Department of Biotechnology, Alagappa University, Karaikudi 630 003, India

² Vision Research Foundation, Sankara Nethralaya, Chennai, Tamil Nadu, India

³ Department of Population and Quantitative Health Sciences, Case Western Reserve University School of Medicine, 1315 Wolstein Research Building, Cleveland, OH 44106, USA

⁴ School of Optometry and Vision Science, University of New South Wales, Sydney, Australia

⁵ Department of Ophthalmology, Harvard Medical School, Massachusetts Eye and Ear Infirmary, Boston, MA, USA

⁶ Singapore Eye Research Institute, Singapore National Eye Centre, Singapore, Republic of Singapore

⁷ Centre for Innovation and Precision Eye Health, Yong Loo Lin School of Medicine, National University of Singapore, Singapore, Republic of Singapore

⁸ Department of Ophthalmology, Yong Loo Lin School of Medicine, National University of Singapore, Singapore, Republic of Singapore

⁹ Ophthalmology and Visual Sciences Academic Clinical Program (Eye ACP), Duke-NUS Medical School, Singapore, Republic of Singapore

¹⁰ Beijing Visual Science and Translational Eye Research Institute, Beijing Tsinghua Changgung Hospital, Tsinghua Medicine, Tsinghua University, Beijing, China

¹¹ Department of Ophthalmology, Flinders Medical and Health Research Institute, Flinders University, Adelaide, SA, Australia

¹² Menzies Institute for Medical Research, University of Tasmania, Hobart, Australia

¹³ MedGenome Labs, Bangalore 560100, India